CHARGER, RADIAC DETECTOR PP-1578( )/PD

1. SCOPE

1.1 This specification covers one type of Radiac Detector Charger designated as PP-1578( )/PD. (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strapping, Flat
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

DRAWINGS

SIGNAL CORPS

SC-A-46439	List of Accessories for Package Tester
SC-DL-146683	Charger-Radiac Detector PP-1578( )/PD

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Charger, Radiac Detector PP-1578( )/PD is a portable, watertight equipment designed to charge Radiacmeters having zeroing voltage requirements between 140 volts and 200 volts which are used by the armed services of the United States, United Kingdom and the Dominion of Canada, such as the following:

UNITED STATES

a. Radiacmeter IM-9( )/PD 0-200mr (milliroentgen)
b. Radiacmeter IM-19( )/PD 0-10r (roentgen)
c. Radiacmeter IM-93( )/UD 0-600r
d. Radiacmeter IM-94( )/PD 0-100r
e. Radiacmeter IM-147( )/PD 0-50r

UNITED KINGDOM

a. No. 1 0-0.5r
b. No. 2A 0-5r
c. No. 3 0-50r
d. No. 4A 0-150r
e. No. 5 0-500r

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The Charger, Radiac Detector PP-1578( )/PD consists of an electrostatic charger, an adapter for United Kingdom radiacmeters, and a carrying case.

3.2 Preproduction samples.- The contractor shall furnish thirty (30) preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3)

3.3 Construction.- The equipment shall be constructed in accordance with Drawing and Data List SC-DL-146683.

3.4 Center electrode.- The height of the center electrode above the bottom of the charging pedestal, at rest, shall be 0.275 ± 0.010 inches. (See 4.6)

3.5 Plastic material and parts.- Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.6 Cleaning.-

3.6.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are installed in the equipment.

3.6.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.7 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.8 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4).

3.8.1 Serial numbers.- Each Charger, Radiac Detector PP-1578( )/PD shall be serial numbered.

3.9 Securing of parts.- The mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to the specified bounce test. Any permanent deformation resulting from the test shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces

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shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single screw will not free the part completely.

3.10 Soldering.-

3.10.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.10.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment.

3.11 Welding.-

3.11.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease, and other foreign material.

3.11.2 Process.- Welds shall have good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.11.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux etc. by suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.12 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.21) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.13 Service conditions.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

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3.13.1 Temperature (see 4.14).-

(a) Operating: Ambient temperature in the range of +125°F to -40°F. (The +125°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.13.2 Relative humidity (see 4.12).- Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for 4 hours for each 24 hour period.

3.13.3 Elevation (see 4.10).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 25,000 feet above sea level.

3.13.4 Immersion (see 4.11).- Three feet of water for 2 hours.

3.13.5 Sand and dust (see 4.13).- As encountered during heavy traffic over dry clay roads or in desert areas.

3.13.6 Bounce.- The equipment shall be capable of meeting full specification performance and shall suffer no physical damage except for minor surface abrasion after being subjected to the test of paragraph 4.9.

3.14 Salt spray.- The equipment shall be capable of withstanding the test specified in paragraph 4.15. The instrument shall not corrode excessively and shall be capable of complying with the requirements of paragraph 3.18 upon completion of this test.

3.15 Weight.- The weight of the equipment, including the carrying case, shall not exceed 12 ounces.

3.16 Force.- The force required to be applied on the center electrode of the charger to close the internal charging switch shall be within appropriate limits as determined by the various Radiacmeters to be charged. The force required shall be a minimum of 6 1/2 pounds and a maximum of 10 pounds. (See 4.7)

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3.17 Viewing.- The equipment shall permit viewing the radiacmeter scale when a radiacmeter is in the charging position and the equipment is pointed toward a light source not exceeding 2 foot-candles. (See 4.20)

3.18 Charging.- The equipment shall charge each type of Radiacmeter listed in 3.1 as follows: At 70°F to 100°F ambient temperature, the charger shall be capable of charging a 600r. radiacmeter (IM-93( )/UD), to zero indication, with no more than 7 complete rotations of the charging knob. Not less than 1/4 complete revolution of the charging knob shall be required to charge a 200mr radiacmeter (IM-9( )/PD) to zero indication. The charging operation shall be smooth and continuous. (See 4.17).

3.19 Voltage requirements.- When rotating the charging knob in the direction of the arrow at approximately 1 revolution per second, there shall be a smooth continuous increase in voltage between 190 and 300 volts. The maximum voltage shall be attained in no more than 7 full turns. (See 4.18).

3.19.1 Voltage limit.- The voltage limiting device shall limit the output voltage to less than 300 volts after initial firing of the neon tubes. (See 4.18.1).

3.20 Polarity.- The equipment shall be capable of permitting reversal of polarity of the voltage applied to the radiacmeter under charge, by reversing the rotation of the control knob. The voltage decrease shall be smooth and continuous. (See 4.19)

3.21 Leakage.- When a 200mr dosimeter is charged to zero and held in the charged position for a period of one minute, the leakage shall not exceed 20mr for this period. (See 4.8)

3.22 Sealing.- The static generator compartment shall be hermetically sealed. (See 4.11)

3.23 Operation.- The equipment shall be capable of meeting the test of 4.23 to indicate that it is operable.

3.24 Life.- After 35,000 knob rotations when tested as specified in paragraph 4.16, the equipment shall still be capable of charging an IM-93( )/PD Radiacmeter, (0-600r/hr.). The equipment shall be considered able to continue to charge if it shall require no more than 12 full turns to charge the IM-93( )/PD to its zero position.

3.25 Technical literature.- Technical literature shall be furnished if specified in the contract. (See 6.2(g))

3.26 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein.

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3.5 General requirements for plastic material and plastic parts
3.6 Cleaning
3.9 General requirements for securing of parts
3.10.1 Flux and cleaning agents for soldering
3.10.2 Process for soldering
3.11.1 Cleaning prior to welding
3.11.2 Process for welding
3.11.3 Cleaning after welding

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.24.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Table II, paragraph 4.5.2 and Table III respectively). The preproduction inspection will normally be performed in this order: (1) bounce and (2) immersion; other preproduction inspection may precede, follow or be interspersed between the foregoing.

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Table I.- Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req Par	Insp Par
Elevation	3.13.3	4.10
Sand and dust	3.13.5	4.13
Bounce	3.13.6	4.9
Salt spray	3.14	4.15
Viewing	3.17	4.20

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.7
Marking	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government except that the operational inspection (4.23) shall be last. (See 6.4)

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Table II.- Group A inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical				
Adapter	3.1	4.22	1% for the entire group	4%
Carrying case	3.1	4.22		
Workmanship	3.26	4.22		
Weight	3.15	4.22	1%	*
Electrode height	3.4	4.6	1%	*
Force	3.16	4.7	1%	*
Electrical				
Charging	3.18	4.17	1% for the entire group	*
Voltage requirements	3.19	4.18		
Voltage limit	3.19.1	4.18.1		
Polarity	3.20	4.19		
Leakage	3.21	4.8		
Operational	3.23	4.23	0.1%	*

* All defects are considered major.

4.5.2 Group B inspection.- This inspection shall consist of the tests specified for immersion in paragraph 4.11 and interchangeability in paragraph 4.21. This inspection, including sampling, shall conform to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection for each inspection position. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.3 Group C inspection.- This inspection shall be as listed in Table III, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

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Table III.- Group C inspection

Inspection	Req Par	Insp Par
Temperature	3.13.1	4.14
Moisture resistance	3.13.2	4.12
Life*	3.24	4.16

*Sample units subjected to this inspection shall not be furnished on contract.

4.5.3.1 Sampling for inspection of equipment.- One sample of Charger, Radiac Detector PP-1578( )/PD, for each group C inspection, shall be selected from each 100 units produced without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will ensure the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.3.)

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Center electrode.- The effective height of the center electrode shall be gaged to determine compliance with the requirement specified in 3.4.

4.7 Force.- The force applied to the center electrode to make electrical contact shall be gaged. The required force shall fall within the limits specified in 3.16.

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4.8 Leakage.- Place a 0-200mr Radiacmeter (IM-9( )/PD) on the charging pedestal and charge to zero. With the radiacmeter held in the charged position for one minute and the generator switch engaged, the unit shall be tested to determine compliance with 3.21.

4.9 Bounce test.- The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its transit case, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 2 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

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4.10 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
3.19	Voltage requirements

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.10(a). The equipment shall meet specified performance for those measurements.

(c) Further reduce the barometric pressure to 11.1 ± 0.1 inches of mercury (simulated 25,000 feet above sea level) and maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. After stabilization, repeat the measurements specified in 4.10(a). The equipment shall meet specified performance for these measurements.

4.11 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ± 2° F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of water leakage. The equipment shall then meet the full specification requirements for the charging, voltage output and polarity tests to determine compliance to the sealing requirement of 3.22.

4.12 Moisture-resistance test.- The equipment shall be tested as follows:

4.12.1 Test conditions.-

(a) It is preferred that the equipment remains in the humidity chamber while the required measurements are being taken. However, if this is not feasible, the equipment may be removed from the chamber for testing.

(b) Complete measurements as rapidly as possible.

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4.12.2 Test method.-

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.12.3.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.12.3 shall be made during the 68° F step as shown on the standard and shall comply with 4.12.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. The equipment shall then be tested and shall meet full specified performance for those measurements specified in 4.12.3.

4.12.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
3.18	Charging	Full specification requirement
3.19	Voltage output	Full specification requirement
3.21	Leakage	Full specification requirement

4.12.4 Failure.- If any equipment fails to meet the performance specified in 4.12.3 during cycling, it does not pass the test.

4.13 Sand and dust test.- The equipment shall be tested as follows:

(a) Place the equipment in its normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 grams per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

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Screen, U. S. Standard Sieve Series	Percent
100-mesh	100
140-mesh	98 ± 2
200-mesh	90 ± 2
325-mesh	75 ± 2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition losses	0 to 2

(d) Maintain internal temperature of test chamber at 77° ± 4° F for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at above conditions, raise the temperature to 160° ± 4° F and maintain it at this value for 6 hours.

(f) Remove equipment from test chamber and allow it to cool to room temperature. The equipment shall meet specified performance for the following measurements:

Paragraph	Measurement
4.17	Charging
4.18	Voltage output
4.19	Polarity

(g) Open the equipment. There shall be no evidence of deterioration of either external or internal parts.

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4.14 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

Step	Paragraph	Measurement	Test results
1, 4, 10	3.18	Charging	12 Turns at Step 4
1, 4, 10	3.19	Voltage requirements	7 Turns at Step 4

4.15 Salt spray test.- The equipment shall be subjected to a 50 hour salt spray test in accordance with the requirements of method 101 of Standard MIL-STD-202 to determine compliance with the requirement of 3.14.

4.16 Life test.- The charging knob shall be rotated under normal conditions in alternate directions at the rate of one rotation per second; forward for 5 seconds and then in reverse for 5 seconds. This operation shall be repeated for a total of 35,000 rotations. After 35,000 rotations, the equipment shall conform to the requirements specified in 3.24.

4.17 Charging.- The radiacmeters, specified in 3.1, shall be placed in the charging position and tested to determine compliance with the requirements of 3.18. The radiac-meters and charger must be in a discharged state prior to use.

4.18 Voltage output.- Place an electrostatic voltmeter, such as Electrostatic Volt-meter, Type 518, as made by the Rawson Electric Instrument Company, or equal, across the output terminals in the charging pedestal. After discharging the voltmeter by placing a shorting bar across either the voltmeter or charger terminals, the charger shall be operated as specified in 3.19.

4.18.1 Voltage limit.- The voltage limiter shall be activated by rotating the charging knob for ten turns. After activation, the unit shall be tested to determine compliance with the requirements specified in 3.19.1.

4.19 Polarity test.- After applying a charge to the electrostatic voltmeter, the rotation of the charging knobs shall be reversed to determine compliance with 3.20.

4.20 Viewing.- The equipment shall be tested, using a standard calibrated light source, to determine compliance with the requirement specified in paragraph 3.17.

4.21 Inspection for dimensional interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) Threaded sections of cap and pedestal.

4.22 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.23 Operational inspection.- The radiacmeters under test, while in no more than a 50 percent discharged state, shall be placed in the charging position and charged to zero. (See 3.23.)

4.24 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.25 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2 (h)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Charging

Voltage requirement

Polarity

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

5.1.1 Level A.- Each Radiac Charger PP-1578( )/PD shall be packaged in accordance with Item 1 through 6 of Figure 1 and Table IV.

5.1.2 Level C.- Each Radiac Charger PP-1578( )/PD shall be packaged in accordance with Items 1 through 6 of Figure 1 and Table IV, except the fiberboard box shall be Class 1 and the tape shall be Class 2 of Specification UU-T-111.

5.1.3 Package Performance.- Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling test and the cyclic exposure test will be required only when invoked in the bid request or contract (See 6.2).

5.2 Packing.-

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BILL OF MATERIAL
TABLE IV

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Bulletin	1	IC-6										
	Bag	1		4	4		MIL-B-117	I	d				
2	Charger, Radiac Detector PP-1578( )/PD w/Adapter in Case	1	III	4	3	3							
3	Cushioning	1		12	7		PPP-P-291	II	2				
4	Box, Folding	1		4-1/2	3-1/2	3-1/2	PPP-B-566	A	B				11
5	Charger PP-1578( )/PD	40											
6	Box, Fiber	1		14	14	12	PPP-B-636	I	2		5	C	RSC
	Tape	6		20	2		PPP-T-76						
	Tape	1		12	2		PPP-T-76						
7	Intermediate Containers	3											
8	Box, Wood or Cleated Plywood	1		38	14-1/2	14-1/2	PPP-B-621		2				4
							PPP-B-601	III	2				A
9	Strapping Metal	3		70	5/8	.015	QQ-S-781	I	2				B

Lev. A - PACKED WEIGHT 160 lbs.
Lev. A - PACKED VOLUME 6 cu. ft.

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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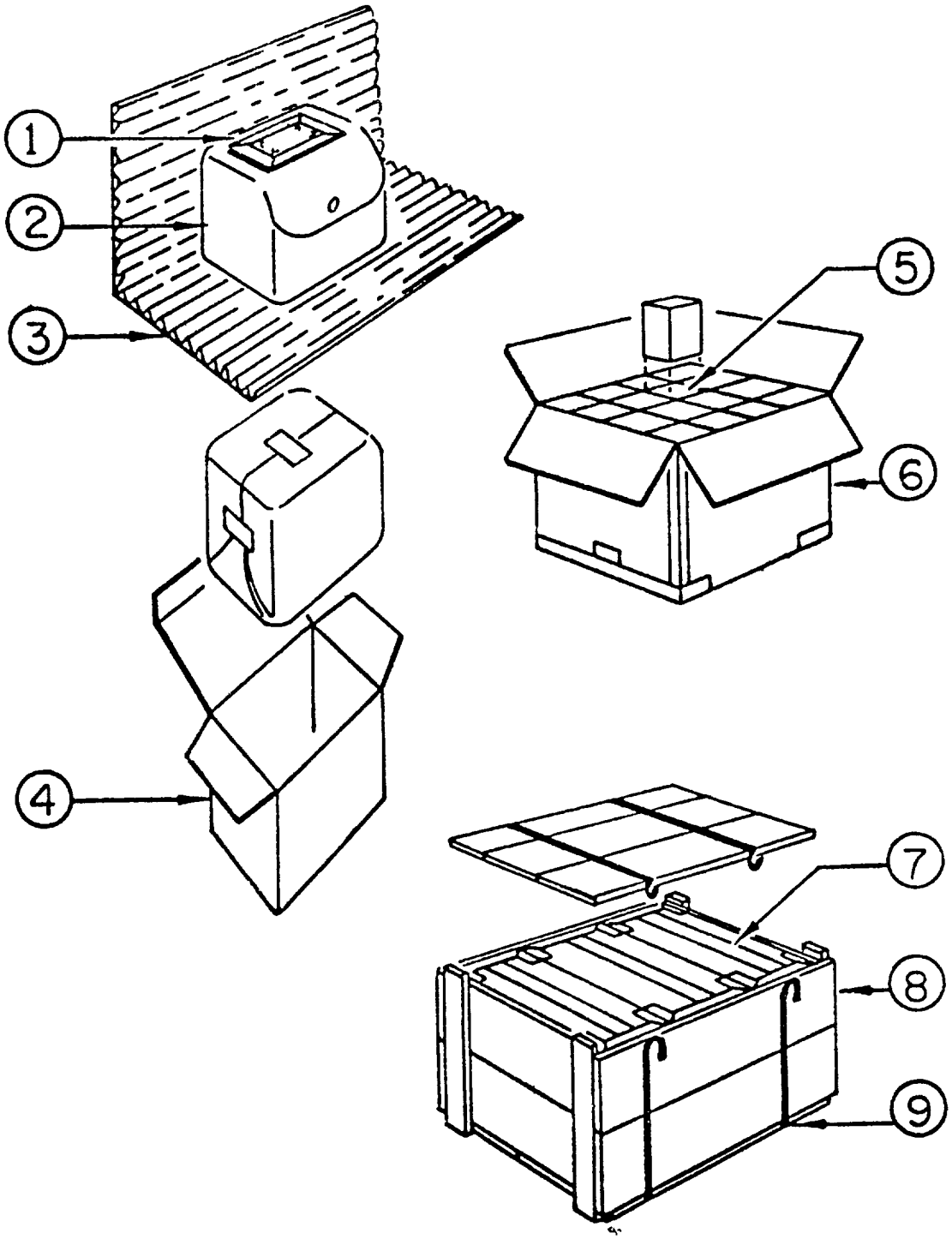


FIGURE 1

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5.2.1 Level A.- Radiac Charger PP-1578( )/PD shall be packed in accordance with Items 7 through 9 of Figure 1 and Table IV.

5.2.2 Level B.- Radiac Charger PP-1578( )/PD shall be packed in accordance with Items 7 and 8 of Figure 1 and Table IV, except the nailed wood box shall be Class 1, or cleated plywood box shall be Type IV and Class 1.

5.2.3 Level C.- Radiac Charger PP-1578( )/PD shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Charger, Radiac Detector PP-1578( )/PD is designed to charge the Radiacmeters of the United States and the United Kingdom as specified in 3.1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.25.)

(h) If rough handling or cyclic exposure test is required.

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6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.4 Location of operational inspection.- It is desirable that the operational inspection (4.23) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-1578W/PD. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.6 Color of textiles.- Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 Dimensional data.- Sizes of packaging materials prescribed in Section 5 are based on the dimensions of the equipment cited on the applicable Bill of Material. When the dimensions of the equipment vary from those cited, the sizes of the packaging materials shall be adjusted accordingly. When shown in the Bill of Materials (see Section 5), corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the fluting used is not the same as that cited in the Bill of Material, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Project Number 4445-A0-25

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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